Decomposition

Decomposition doesn't sound like the sort of thing anyone would want happening near a computer. Luckily, this decomposition is actually the first step in the computational thinking process.

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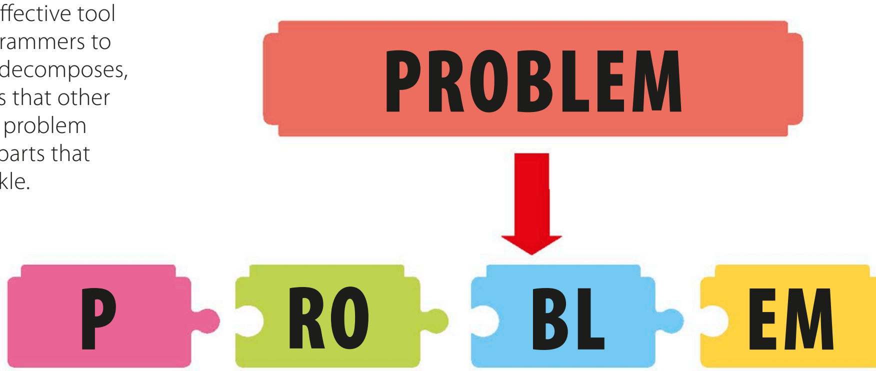
What is decomposition?

Decomposition is the process of breaking down problems into smaller components. An effective tool in computational thinking, it allows programmers to build effective solutions. When an apple decomposes, it's breaking down into simpler chemicals that other plants can use as food. In a similar way, a problem can be solved by splitting it into smaller parts that a programmer already knows how to tackle.



▷ **Find the sub-problems**
A lot of everyday problems are actually made up of smaller parts, which we can call sub-problems.

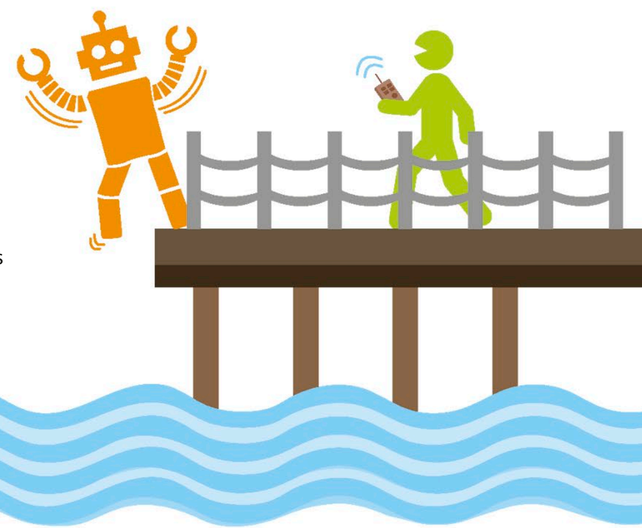
LINGO

Modular code

Building a program by writing small amounts of code to solve sub-problems is known as a modular approach. If there is a problem with a part of the code, it can easily be taken out and fixed. Each smaller solution is tested before it's added to the main program. Breaking the original problem into sub-problems also gives programmers the option of sharing the work among a team.

Computer sense

Computers, unlike people, don't have any common sense or knowledge of how things work. They do exactly what they're told to do, even if the instructions are ridiculous or totally wrong. When writing a program for a computer to solve a problem, computer scientists must include precise and detailed instructions on how to do each tiny step.



▷ **Task gone wrong**
If a computer is given instructions that are inaccurate, in the wrong order, or incomplete, it won't complete the task successfully.